NATURAL LANGUAGE PROCESSING

(NLP)

What is natural language processing?

NLP: is a part of computer science and AI which deals with human languages.

The Internet is full of text data. The oral and written communication between humans are done through languages.

Applications of NLP

By utilizing NLP, one can solve a wide range of problems such as : Spell checking, Keyword search, Information extraction from websites, named entity recognition, summarization, Sentiment analysis , Speech recognition, Voice assistants like the Siri, Google assistant, and Cortana, Chatbot, Machine translation, Social media monitoring, Generating responses, Making appointments

NLP components

- **NLU** (Natural Language Understanding)
- **NLG** (Natural Language Generation): The process of producing meaningful phrases and sentences in the form of natural language.

NLU involves mapping an input in natural language into useful representation and analyzing different aspects of the language. NLG involves text planning and sentence planning.

* NLU is normally considered much harder than NLG.

NLU Ambiguity

Lexical ambiguity (semantic ambiguity)

The presence of two or more possible meanings for a single word

E.g. The fisherman went to the bank.

Syntactic ambiguity (structural or grammatical ambiguity)

The presence of two or more possible meanings within a single sentence or a sequence of words

E.g. The chicken is ready to eat.
I saw the man with the binoculars.
Visiting relatives can be boring.

Referential ambiguity

This kind arises when we refer to something using pronouns

E.g. Ahmad helped Rami. He is a nice guy.

Tokenization

- A process of breaking up strings into tokens which in turn are small structures or units that can be used for tokenization.

The N-gram model

- It is a model that assigns probabilities to sentences and sequences of words.

- Think of an N-gram as the sequence of N words, by that notion, a 2-gram (or bigram) is a two-word sequence of words, and a 3-gram (or trigram) is a three-word sequence of words.

- **E.g.1:** I want to learn Machine Learning

Unigram:
[I][want][to][learn][Machine][Learning]

Biagram:
[I want][want to][to learn][learn Machine][Machine Learning]

Trigram:
[I want to][want to learn][to learn Machine][learn Machine Learning]

- You can also generate for N=4,5,6

Note:

If X=Number of words in a given sentence, the number of N-grams for sentence would be:

The number of N-grams = X - (N - 1)

- **E.g.2:** Let me explain with an example

Unigram - [Let] [me] [explain] [with] [an] [example]

Bigram [let me] [me explain] [explain with] [with an] [an example]

Trigram [let me explain] [me explain with] [explain with an] [with an example]